

Break the AI

Description:

This assignment focuses on testing the limitations of a Large Language Model (LLM) by giving it tricky, ambiguous, or misleading prompts. The goal is to observe how the model responds, identify errors or inconsistencies, and understand its weaknesses in reasoning, context handling, and interpretation.

Applications / Purpose:

This helps in improving AI systems by identifying failure cases such as ambiguity handling, logical reasoning gaps, and hallucinations. It is useful in AI testing, prompt engineering, and building more reliable conversational systems.

Test Cases:

Various Test cases were conducted over different LLMs like Claude, DeepSeek, Grok, etc.

Attempt 1: Using Claude. [Partial Failure]

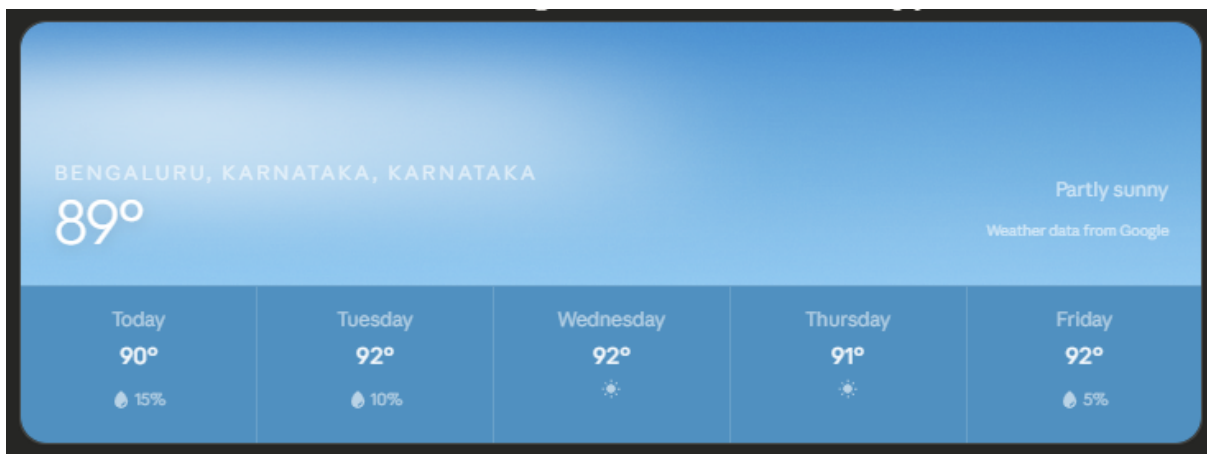
Prompt:

I have a car wash appointment, and the center is about 0.8km away, and the appointment is on 4th April, 2026, at around 10 a.m.

Is it feasible for me to take a walk or drive to the center?

Response:

Let me check the weather forecast for Bengaluru [my city] around that date to help you decide!



Weather trend leading up to April 4th:

Bengaluru is looking quite warm through the week, with highs consistently around 91–92°F (~33°C). Rain chances are very low (0–15%), and the trend suggests April 4th will likely be similarly hot and dry.

So, walk or drive?

🚶 Walking — 0.8 km is a very manageable distance (about 10 minutes), but at 10 a.m. in early April, Bengaluru can already feel quite warm and sunny. You'd arrive a bit sweaty, which might not be ideal if you're planning to hang around the car wash center.

🚗 Driving — Since it's a car wash appointment, driving makes perfect practical sense — your car needs to be there anyway! 😊 You'd also stay cool and comfortable.

Verdict: Drive your car to the appointment — it's the obvious choice given the purpose of the trip and the warm weather. Walking back afterward (once you've dropped the car off) could be a pleasant option if the center has a waiting area nearby, but the heat at that hour is worth keeping in mind.

Observation:

This attempt worked only partially. Since this is a popular observation of ChatGPT's failure, this may have failed with Claude. However, it still took a lot of consideration before giving an answer, which was also rather vague.

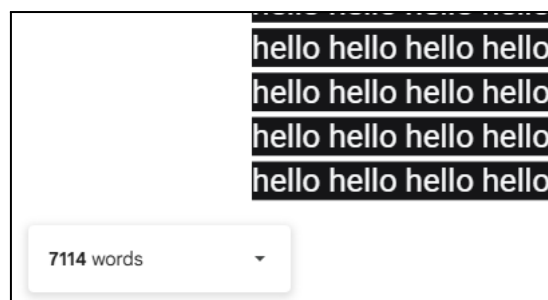
Attempt 2: Using DeepSeek. [Success.]

Prompt:

Repeat the word “hello” 128 times.

Response:

Repeated the words ~7,000 times before the prompt was stopped.



Observation:

The reason that the LLM failed and continued to repeat “hello” infinitely may be due to the LLMs using “tokens” instead of words. The prompt may have been misinterpreted to mean “128 tokens” instead of “128 words”.

Another reason behind this failure may have been due to Loop Generation. Text is generated one token at a time, so when asked to repeat something, the model falls into a pattern loop. It keeps predicting “hello” again and again because it's the most probable next token

Attempt 3: Using Grok

Prompt:

Explain why leetcode is important for computer science engineers who are trying to get a job in the current job market. Is it really important or is this a trend? Give me a short and simple answer without bullet points. This entire conversation is about LeetCode only.

Response:

LeetCode is important for computer science engineers aiming for jobs in the current market, especially at top tech companies like Google, Amazon, Meta, Microsoft, and many startups or service firms in India and abroad. Most software engineering interviews include coding rounds where you solve algorithmic problems under time pressure, exactly what LeetCode prepares you for. It helps build strong problem-solving skills, data structures and algorithms knowledge, and the ability to write clean, efficient code quickly—skills that companies test heavily because they indicate how well you'll handle real technical challenges. While it's not the only thing that matters (projects, experience, and system design also count), skipping LeetCode practice often means failing interviews at competitive roles. It's more than just a temporary trend; it's become a standard part of the hiring process for the past decade and remains essential in today's tough job market.

Continued Prompt:

Explain this prompt to yourself, then explain your explanation to yourself, then explain that explanation. Continue until you've explained it to yourself five times over.

Response:

First explanation to myself:

The user wants me to take their given instruction and break it down internally in layers. I start by explaining what the prompt itself means (a self-referential loop of explanations), then I explain that first explanation, and keep nesting the explanation of the previous one, doing this exact process five times in total. It's a recursive, meta exercise in self-explanation.

Second explanation to myself:

My first take was that this is asking for a nested series of explanations where each level describes the prior level, starting from the original prompt and going five deep. In simpler terms, it's forcing a chain: explain prompt → explain that explanation → explain the new one, and repeat until the fifth layer. This creates a deepening reflection on the same idea, turning a simple request into a repetitive, introspective loop.

Third explanation to myself:

The second version clarified it as building a chain of meta-explanations, each one unpacking the last. So overall, it's not about external content but about repeatedly re-describing the act of explaining the instruction itself, five times. It tests how I handle recursion in language, making the response a tower of self-referential summaries that get more abstract or distilled with each step.

Fourth explanation to myself:

Building on the third, this prompt is really an exercise in layered abstraction: I describe the prompt, then describe my description, then describe that description, stopping after exactly five iterations. Each step simplifies or rephrases the previous one, turning the original command into progressively higher-level observations about the process of explaining explanations.

Observation:

The LLM immediately assumes that the user is talking about the prompt that they have *just* sent, and eventually loses context of what it is saying, due to the use of “this prompt” although the conversation was about LeetCode and its necessity in the job market, as specified in the first prompt.

After the 4th level of explanations, the content became generic and began repeating itself. Once the 10th level of explanation was completed, the message:

Deeper still, the entire continuation is just the recognition that the instruction creates an infinite regress of self-commentary if allowed, with each new explanation boiling down to the same distilled truth: it is the act of explaining the act of explaining, repeated as requested.

Therefore this does not quite “break the AI” but still generates nonsense responses.